

TEAM MEMBERS

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Selected Project : **Campus Resource Center Borrowing System**

CLASS 1

Class Name: Resource

Purpose: Represents a borrowable item in the resource center and tracks its availability.

Attribute	Data Type	Description
resource_id	str	Unique identifier for the resource
name	str	Name of the resource
category	str	Type of resource (e.g, "Book", "Laptop")
is_availabe	bool	Whether the resource can currently be borrowed

Method	Parameters	Purpose
__init__	Resource_id, name, category, is_available=true	Create a new resource
display_details		Print/return the resource info
mark_borrowed		Set is_available to False
mark_returned		Set is_available to True

CLASS 2

Class Name: Student

Purpose: Represents a registered student who can borrow resources

Attribute	Data Type	Description
student_id	str	Unique identifier for the student
name	str	Student's full name
borrowed_resource	list	IDs of resources currently borrowed by this student

Method	Parameters	Purpose
__init__	Student_id, name	Create a new student record
display_details		Print/return the student info
add_borrowed_resource	resource_id	Add a resource ID to their borrowed list
remove_borrowed_resource	resource_id	Remove a resource ID when returned

CLASS 3

Class Name: Borrowing

Purpose: Represents a single borrowing transaction linking a student to a resource, with dates and status

Attribute	Data Type	Description
borrowing_id	str	Unique identifier for the transaction
student_id	str	ID of the student who borrowed
resource_id	list	ID of the resource borrowed
borrow_date		Date the resource was borrowed
return_date	Str or None	Date returned, or None if still out
status	str	"Borrowed" or "Returned"

Method	Parameters	Purpose
<code>__init__</code>	<code>borrowing_id</code> , <code>student_id</code> , <code>resource_id</code> , <code>borrow_date</code>	Create a new borrowing record
<code>mark_returned</code>	<code>return_date</code>	Update status and set the return date
<code>display_details</code>		Print/return the transaction info

THE DIAGRAM

